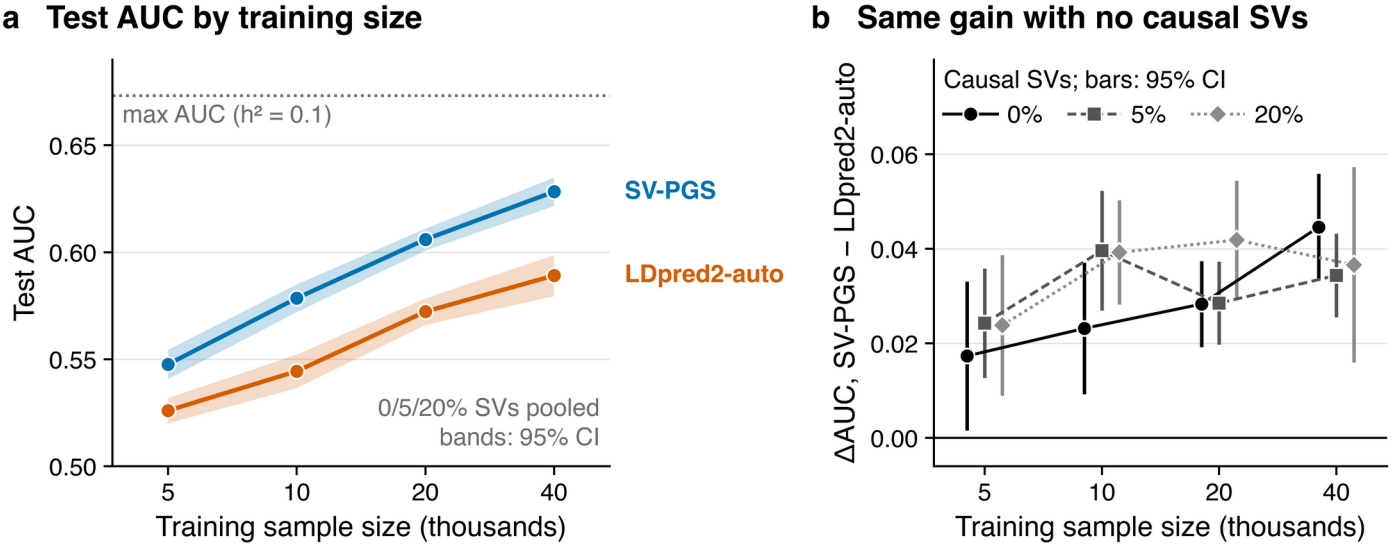



Figure 1 | SV-PGS is more accurate than LDpred2-auto at every training size, by the same margin whether 0%, 5% or 20% of the causal variants are SVs, so the gain does not come from causal SVs. A simulated binary trait on chromosome 22 (liability $h^2$ = 0.1, prevalence 10%, 200 causal variants) scored in the same 10,000 held-out people; SV-PGS fits every SNV, indel and imputed SV, LDpred2-auto HapMap3-density SNVs of the same training people. a, Test AUC, mean and Student-t 95% interval over the three settings' replicates pooled (42, 37, 38, 16 at 5k-40k); dotted, the largest AUC any score can reach, that of the true genetic value at $h^2$ = 0.1 (Wray et al. 2010). b, Paired difference in AUC on the same held-out people, per setting, mean and Student-t 95% interval.

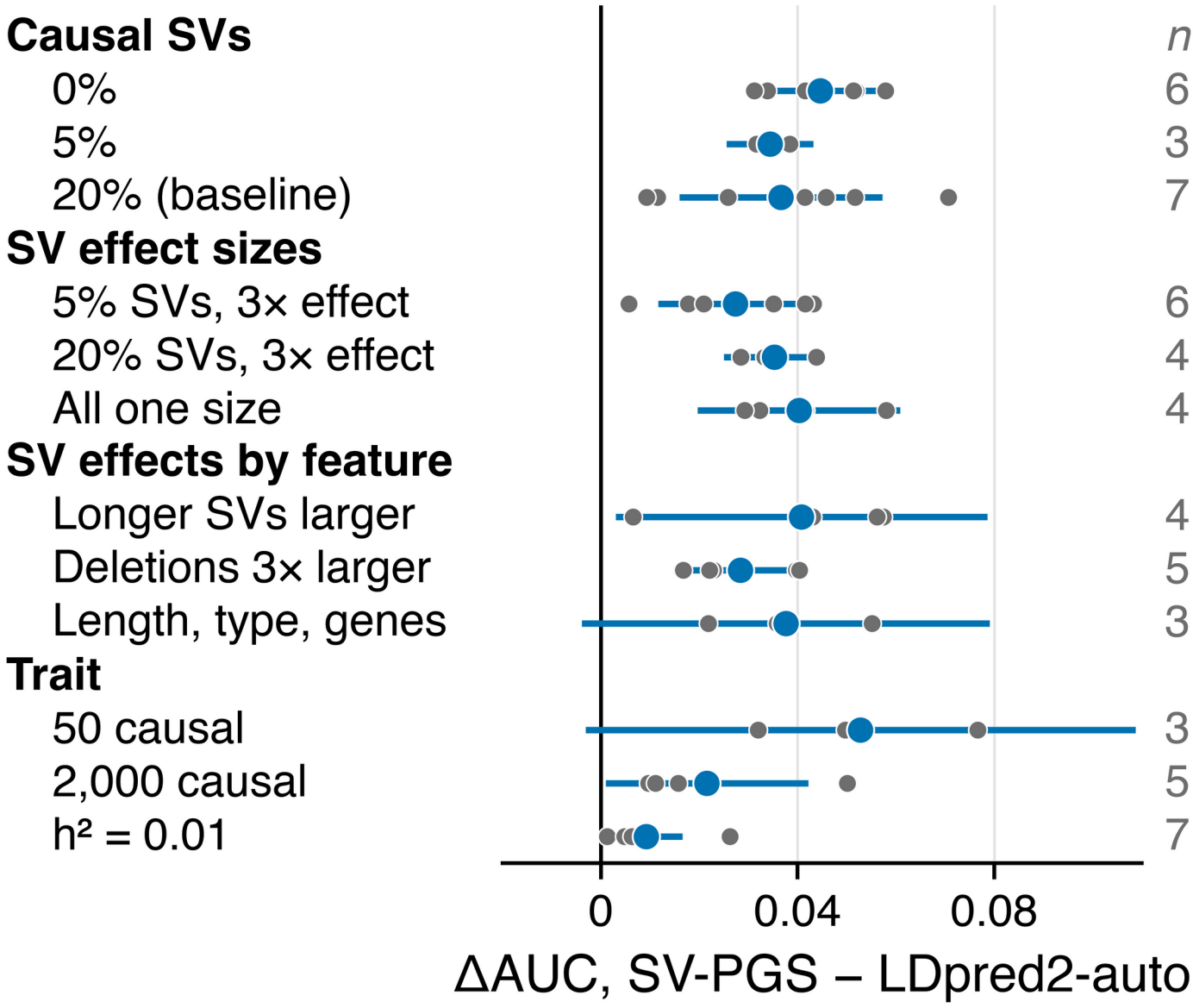



Figure 2 | SV-PGS is more accurate than LDpred2-auto across simulated genetic architectures. Paired difference in test AUC on the same 10,000 held-out people between SV-PGS (every SNV, indel and imputed SV) and LDpred2-auto (HapMap3-density SNVs), both trained on the same 40,000 people: replicates (grey), mean and Student-t 95% interval (blue; an arrowhead marks one that runs past 0.11), n replicates at right. SV-PGS had the higher AUC in 57 of 57 replicates. The baseline architecture has liability $h^2 = 0.1$, prevalence 10%, 200 causal variants and 20% causal SVs with SV effects drawn like small-variant effects, and every other changes one thing of it: '3×' triples the SV effect SD, 'a few very large' draws SV effects from a Student t with 3 degrees of freedom, 'all one size' gives every causal SV the same effect size with a random sign, 'well-imputed' draws causal SVs only from those imputed with $r^2 \geq 0.8$, and 'common' only from SVs of frequency ≥ 5%. Not yet replicated three times at 40,000: a few very large SV effects; only well-imputed SVs; only well-imputed SVs, 3× effect; only common SVs; prevalence 1%.

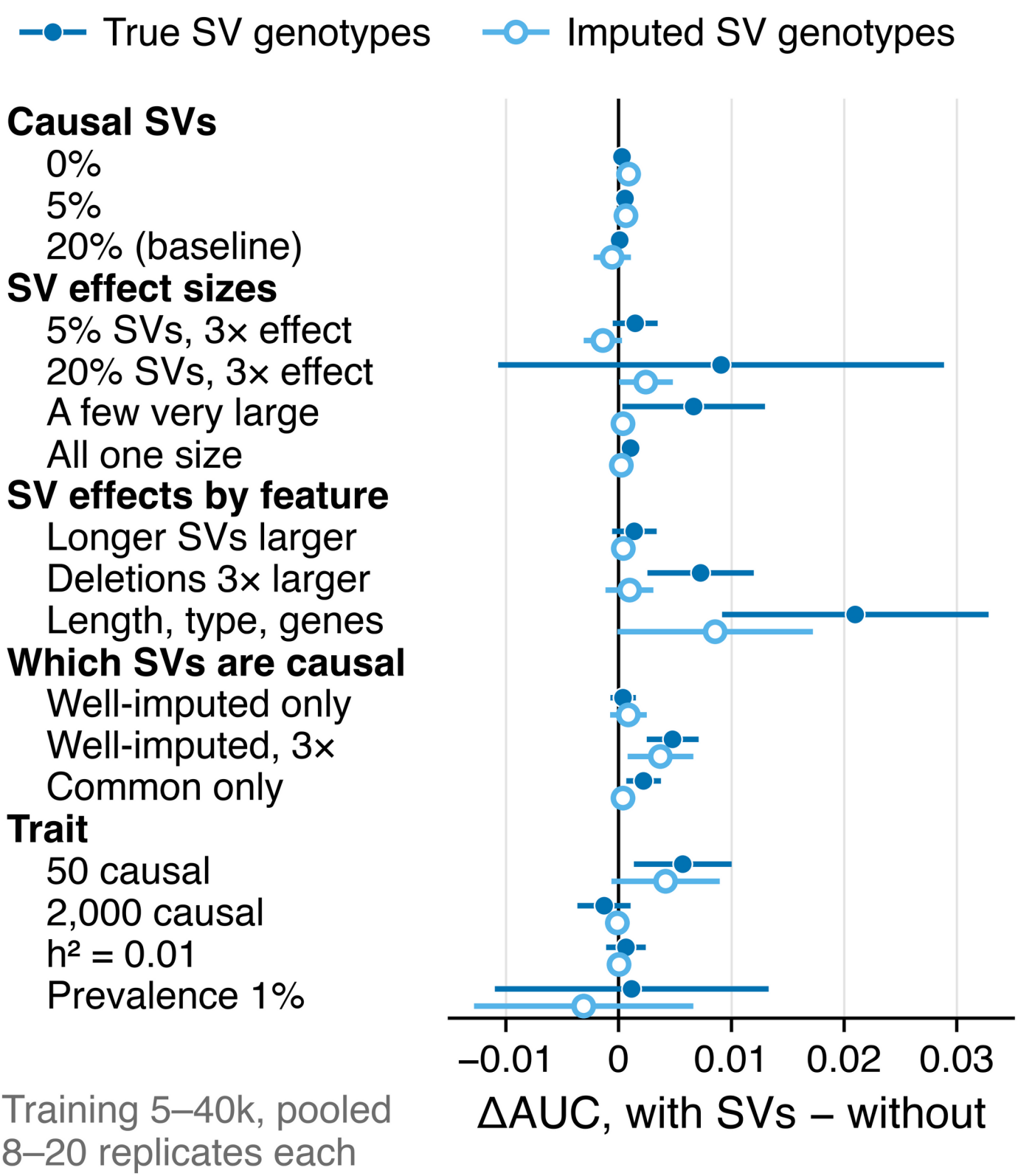



Figure 3 | SVs add accuracy when their effects are large and they are measured well; imputation removes most of the gain. Paired difference in test AUC on the same held-out people between SV-PGS fitted with and without SVs on one 3 Mb chromosome 22 window (60,000 records), with every SV measured by its true genotype (filled) or imputed by Beagle from SNV haplotypes (open): mean and Student-t 95% interval over 8-20 replicates, every training size from 5,000 to 40,000 pooled (Figure 4 shows them apart). The true-genotype interval lies above zero for 7 of 17 architectures. Same simulated people, truth and fits; only the SV genotypes differ. The widest true-genotype interval, 20% SVs, 3× effect, comes from one replicate at –0.085. The baseline architecture has liability $h^2 = 0.1$, prevalence 10%, 200 causal variants and 20% causal SVs with SV effects drawn like small-variant effects, and every other changes one thing of it: '3×' triples the SV effect SD, 'a few very large' draws SV effects from a Student t with 3 degrees of freedom, 'all one size' gives every causal SV the same effect size with a random sign, 'well-imputed' draws causal SVs only from those imputed with $r^2 \geq 0.8$, and 'common' only from SVs of frequency ≥ 5%.

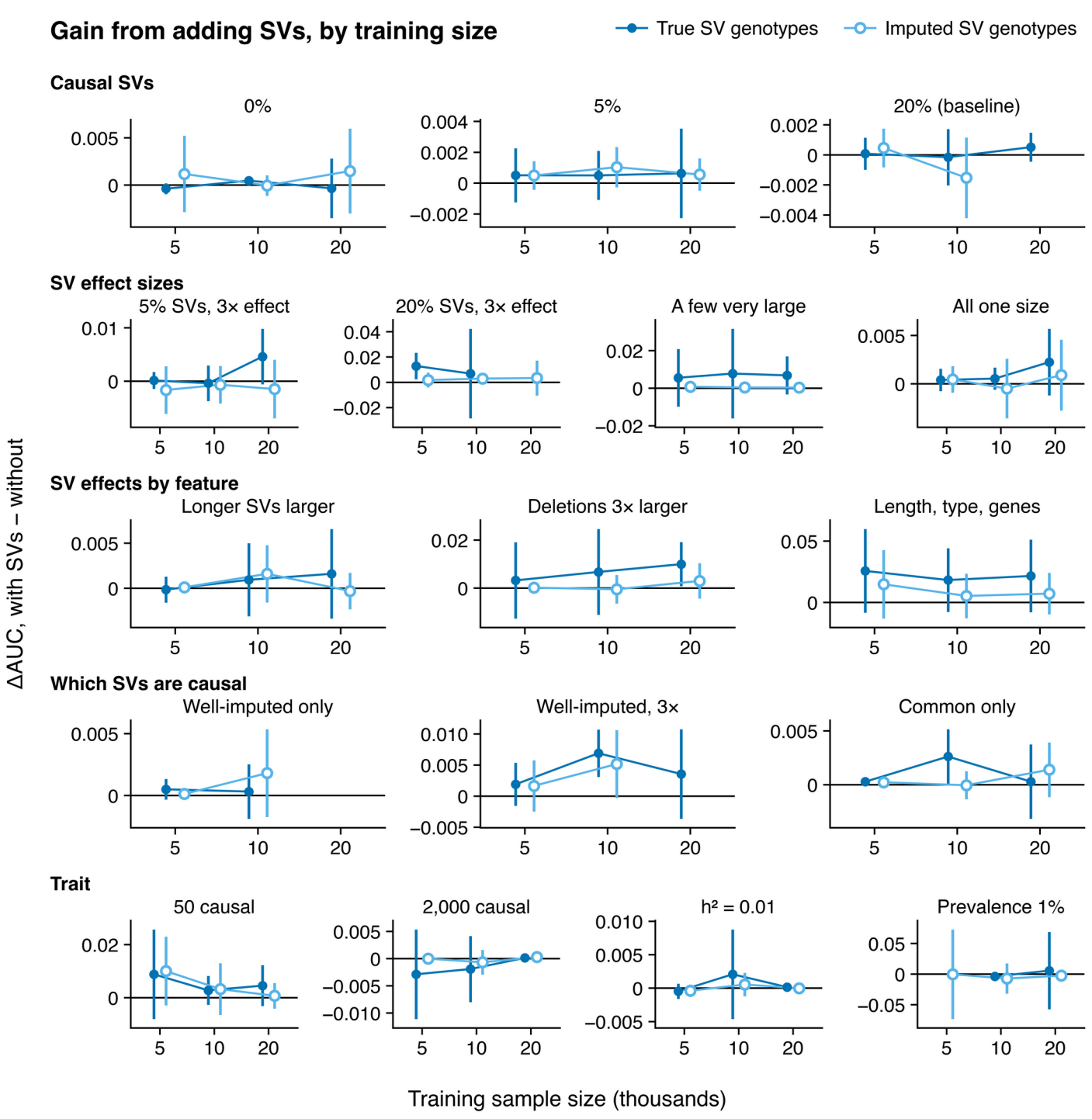



Figure 4 | The gain from adding SVs at each training size from 5,000 to 20,000, for each simulated architecture. Each panel shows the paired difference in test AUC between SV-PGS fitted with and without SVs on one 3 Mb chromosome 22 window, with SVs measured by their true genotypes (filled) or imputed by Beagle (open): the mean over replicates and its Student-t 95% interval. Each panel has its own y-axis scale. A mean is drawn when its training size has at least three replicates. Each replicate uses the same simulated truth and the same test people at every training size. We fitted 0-3 replicates per architecture and genotype source at 40,000 training people, too few to show; Figure 3 pools them with the other sizes. The baseline architecture has liability $h^2 = 0.1$, prevalence 10%, 200 causal variants and 20% causal SVs with SV effects drawn like small-variant effects, and every other changes one thing of it: '3×' triples the SV effect SD, 'a few very large' draws SV effects from a Student t with 3 degrees of freedom, 'all one size' gives every causal SV the same effect size with a random sign, 'well-imputed' draws causal SVs only from those imputed with $r^2 \geq 0.8$, and 'common' only from SVs of frequency ≥ 5%.

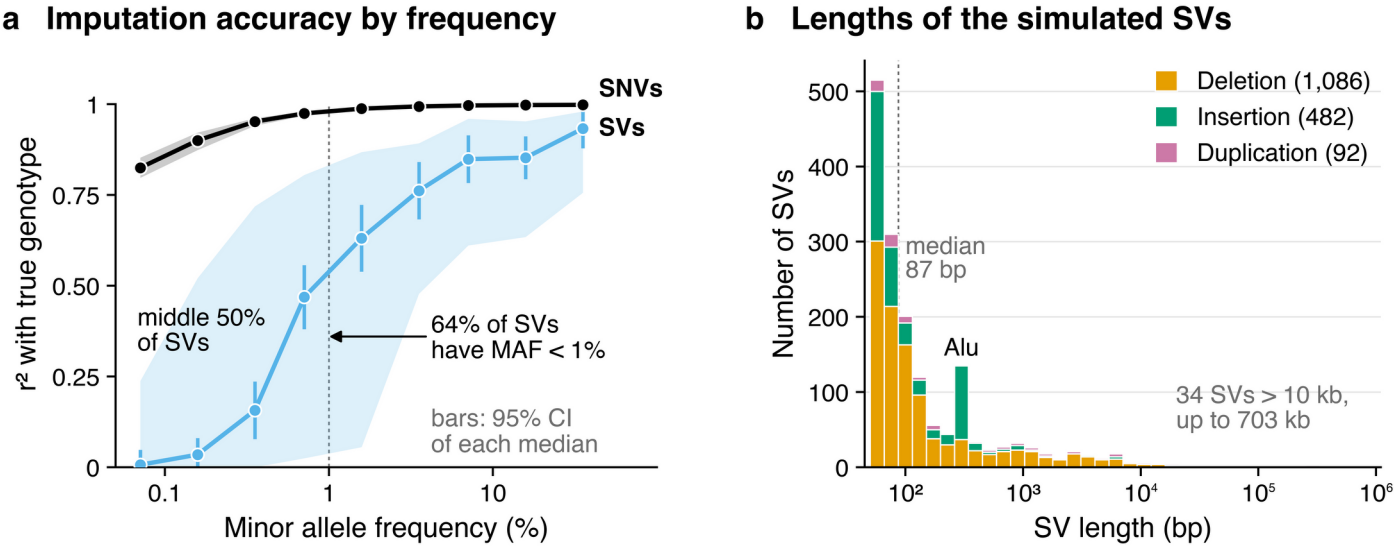



Figure 5 | The simulated SVs are mostly rare and short, and rare SVs impute poorly. a, The $r^2$ of each variant's Beagle-imputed dosage with its true genotype over all 50,000 simulated people (not Beagle's DR2), for 1,662 SVs and 6,000 sampled SNVs on chromosome 22, by minor allele frequency: each bin's median (points) with an approximate 95% interval (1.82 IQR/$\sqrt{n}$; bars), over the bin's middle 50% of variants (bands; SNVs grey, SVs blue). b, Lengths of the 1,662 measured SVs, stacked by class (records without a type are classed by the sign of their length change); the insertions' peak near 300 bp is where Alu elements lie. bench-sim v7 cohort from public 1000 Genomes haplotypes.

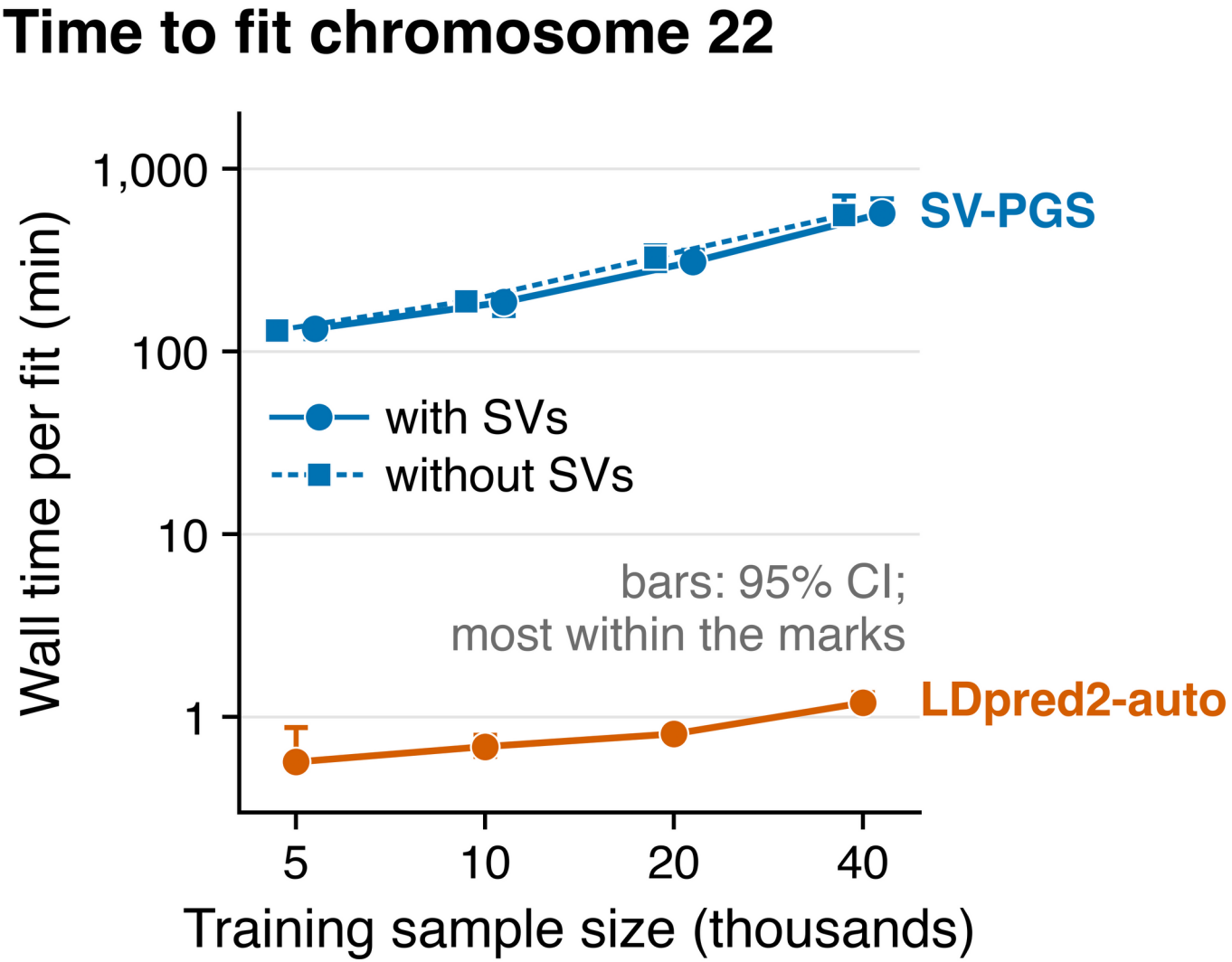



Figure 6 | SV-PGS fits take hours and LDpred2-auto's minutes; SVs add no time. Wall time of one fit on chromosome 22 per method and training size, pooled over every architecture: median and its distribution-free 95% interval (59-87 fits each; capped bars, most within the marks). MSI Slurm jobs, one fit per job; SV-PGS fits every variant by empirical Bayes (circles and solid line, with SVs; squares and dashed line, without), LDpred2-auto HapMap3-density SNVs. The results do not record each job's hardware, so the gap compares the methods as they were run, not on equal resources.